Senzing

Bootcamp

Completion Recap

A record of everything you built and learned, module by module. Keep it, revisit it, and share it with your team.

BOOTCAMPER
Senzing Bootcamper

STARTED
2026-11-11T10:13:40-05:00

COMPLETED
2026-11-12

PROGRAMMING LANGUAGE
Java

PATH
Core

PLUGIN VERSION
0.6.1

Modules in this recap

- Entity Resolution Concepts
- Discover the Business Problem
- SDK setup
- System verification
- Truth Set visualization
- Data collection
- Data Quality, Mapping, and Transformation
- Data processing
- Query, Visualize and Discover

Run environment

Operating system: macOS 26.6.2 (arm64)

Python version: Python 3.9.6

Language runtime: Java 25.0.2

Senzing SDK: 4.3.2

Database: SQLite

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Entity Resolution Concepts

Completed November 11, 2026

Information Shared

- Entity resolution determines whether records refer to the same entity, are related, or do not match. False negatives split records for the same entity; false positives merge records for different entities.
- Ambiguous records can be held as possible matches until distinguishing information arrives, avoiding an arbitrary assignment that could create an invisible false positive.
- Senzing documentation describes resolving incoming records against known entities using principles based on expected attribute behaviors; common behaviors for people and organizations are preconfigured, and exceptions can prompt earlier decisions to be revisited.
- Disclosed relationships are explicitly stated; discovered relationships can be inferred from shared attributes. Senzing documentation describes explanations for why, why not, and how matching decisions were made.
- Scalability should be evaluated against current and future data volumes and complexity.

Questions & Responses

- **Q:** Do you have any questions about entity resolution before we continue?
  - **R:** No.
- **Q:** Would you like a few quick questions to help the concepts stick?
  - **R:** Yes.
- **Q:** A record could plausibly belong to more than one entity. Before new information distinguishes them, what should a well-designed system do?
  - **R:** 2 - hold it as a possible match.
- **Q:** Why does a principle-based approach treat identifiers and dates of birth differently?
  - **R:** 2 - an identifier may point to one person, while a date of birth can be shared by many.
- **Q:** Which example is a discovered relationship?
  - **R:** 2 - records share an address, suggesting a relationship.
- **Q:** When resolving a new record, which approach uses what is already known about an entity?
  - **R:** 2 - compare it with information gathered across the entity's records.
- **Q:** Are you ready to move on to the next module: Discover the Business Problem?
  - **R:** Yes.

Actions Taken

- Presented the entity-resolution primer and optional knowledge check, using Senzing documentation retrieved through MCP.
- Completed four multiple-choice knowledge-check questions.
- No source data or application code was created during this conceptual primer.

End-of-Module Summary

What you accomplished:

- Learned the difference between false positives, false negatives, and ambiguous matches.
- Explored principle-based matching and disclosed and discovered relationships.

Files produced:

- (no files - conceptual primer)

Why it matters:

These concepts provide context for defining an entity-resolution business problem and making informed data and mapping decisions later in the bootcamp.

Discover the Business Problem

Completed November 11, 2026

Information Shared

- Reminded the bootcamper to use data only with permission and to consider anonymizing PII for testing.
- Reviewed a business-problem pattern gallery; the bootcamper said none matched, then provided a real business-case brief.
- Summarized the fictional insurance claims-fraud case: policyholders, claim-paid vendors, and an SIU watchlist across three sources totaling 159 synthetic records.
- Reviewed Senzing documentation describing entity matching, relationship discovery and management, revisiting previous decisions, and risk/fraud and customer-data-management use cases.
- Clarified that the draft requires reviewer approval before its data may be loaded.

Questions & Responses

- **Q:** Would you like to see examples of common business problems that entity resolution can solve?
  - **R:** Yes.
- **Q:** Do any of these patterns match your situation?
  - **R:** No.
- **Q:** How would you like to define the business problem?
  - **R:** The bootcamper provided a local business-case brief.
- **Q:** Does that summary capture your situation accurately?
  - **R:** Yes.
- **Q:** What does the end result look like?
  - **R:** A clean master list, an API, and reports.
- **Q:** Do you have any diagrams of your data architecture and flows you'd like to share?
  - **R:** No.
- **Q:** Will your entity-resolution results need to interface with other software (CRM, search engine, data warehouse, API gateway, downstream app)?
  - **R:** No.
- **Q:** Where do you plan to deploy the final solution?
  - **R:** 3 (Local / on-premises).
- **Q:** Does this accurately capture your problem and approach?
  - **R:** Yes.

Actions Taken

- Created docs/business_problem.md with the confirmed scenario, data sources, matching criteria, success outcomes, desired outputs, and unresolved items.
- Updated the Overview and Business Problem sections in README.md.
- Recorded integration_targets: [] and deployment_target: local in config/bootcamp_preferences.yaml.

- Created docs/stakeholder_summary_module1.md with the proposed approach, sources, key findings, next steps, and ROI considerations.
- No source data was loaded; the draft remains subject to reviewer approval.

End-of-Module Summary

What you accomplished:

- Defined and confirmed an insurance claims-fraud business problem using the provided synthetic-data brief.
- Selected a clean master list, an API, and reports, and recorded a local / on-premises deployment target with no downstream integrations.
- Identified the unresolved measures and operational details to establish in later modules.

Files produced:

- docs/business_problem.md - confirmed business problem and requirements.
- README.md - project overview and business problem.
- docs/stakeholder_summary_module1.md - stakeholder-facing summary and next steps.
- docs/bootcamp_recap.md - bootcamp recap with this module's results.

Why it matters:

This confirmed definition guides the next stages of source approval, data preparation, processing, and validation. The synthetic data remains unloaded until a reviewer approves the draft.

SDK setup

Completed November 11, 2026

Information Shared

- The existing Senzing SDK was reused; its version query reported 4.3.2.
- The Java binding loaded from the installed sz-sdk.jar; the project-local environment and launcher avoid changes to global shell configuration.
- The Homebrew update check could not complete because /opt/homebrew/Cellar is not writable, so whether a newer version is available remains unknown.
- The provisional license reading was 500 records, and the post-configuration measurement remained 500.
- SQLite was selected for the project. Its schema and default Senzing configuration were created, and an engine call succeeded.
- The insurance case data remains unloaded pending reviewer approval.

Questions & Responses

- **Q:** Which database would you like to use?
  - **R:** 1 - SQLite.

Actions Taken

- Created and validated src/scripts/senzing-env.sh and src/scripts/run-java.sh.
- Fetched, hash-verified, compiled, and ran the official Java version and license examples.
- Created database/G2C.db, applied the installed SQLite schema, and wrote config/engine_config.json using the platform paths and the project's absolute database path.
- Seeded the default configuration, reinitialized the environment, and successfully ran the engine-priming check.
- Saved the measured license response to config/license.json and created the project directory layout.
- Did not load any insurance case records.

End-of-Module Summary

What you accomplished:

- Reused and verified the installed Senzing SDK with Java.
- Configured the project-local SQLite database and default configuration, then confirmed an engine call succeeds.

Files produced:

- src/scripts/senzing-env.sh - project-local SDK environment.
- src/scripts/run-java.sh - Java launcher with the native library paths.
- src/java/information/GetVersion.java and src/java/information/GetLicense.java - SDK version and license

checks.

- src/java/configuration/InitDefaultConfig.java - default configuration seeding.
- src/java/initialization/EnginePriming.java - engine connection check.
- config/engine_config.json - project engine settings.
- config/license.json - measured license response.
- database/G2C.db - schema-initialized SQLite repository.
- config/bootcamp_preferences.yaml and config/bootcamp_progress.json - database choice and bootcamp state.
- docs/bootcamp_recap.md and docs/video/broll.json - module recap and video manifest.

Why it matters:

The project now has a verified local engine and persistent repository for the next modules, while the case data stays unloaded until reviewer approval.

System verification

Completed November 11, 2026

Information Shared

- The system was verified end to end using only four generated synthetic records; the insurance-case data remains unloaded pending reviewer approval.
- Results validation is reported separately from installation checks because it tests an expected resolution outcome, not whether the installation works.
- The Truth Set visualization is a separate module that follows System verification.

Questions & Responses

- {none this module}

Actions Taken

- Confirmed MCP connectivity and engine initialization, then generated and ran Java SDK initialization, loading, resolution, and database-operation programs.
- Registered the test-only VERIFY source and loaded four synthetic records.
- Confirmed the records resolved into two entities: a three-record cluster and a distinct singleton.
- Verified the loaded count, read the merge entity by ID, and found it through a name-and-address search.
- Deleted the four synthetic records through the SDK and verified none remained.

End-of-Module Summary

What you accomplished:

- Verified Java SDK initialization, compilation, data loading, entity resolution, read, and search operations using four synthetic records.
- Confirmed the expected three-record cluster and singleton, then removed all verification records from the database.

Files produced:

- src/system_verification/verification_data.jsonl - synthetic verification records.
- src/system_verification/verify_init.java - SDK initialization check.
- src/system_verification/verify_pipeline.java - file-driven data loading pipeline.
- src/system_verification/register_data_sources.java - test-source registration program.
- src/system_verification/verify_resolution.java - entity-resolution validation.
- src/system_verification/verify_database_operations.java - write-count, read, and search checks.
- src/system_verification/purge_verification_data.java - synthetic-data cleanup and verification.
- config/data_sources.yaml - marks VERIFY as test-only.

Why it matters:

The installation is verified without loading the insurance-case data, and the test records have been removed so the database is ready for the next module.

Truth Set visualization

Completed November 11, 2026

Information Shared

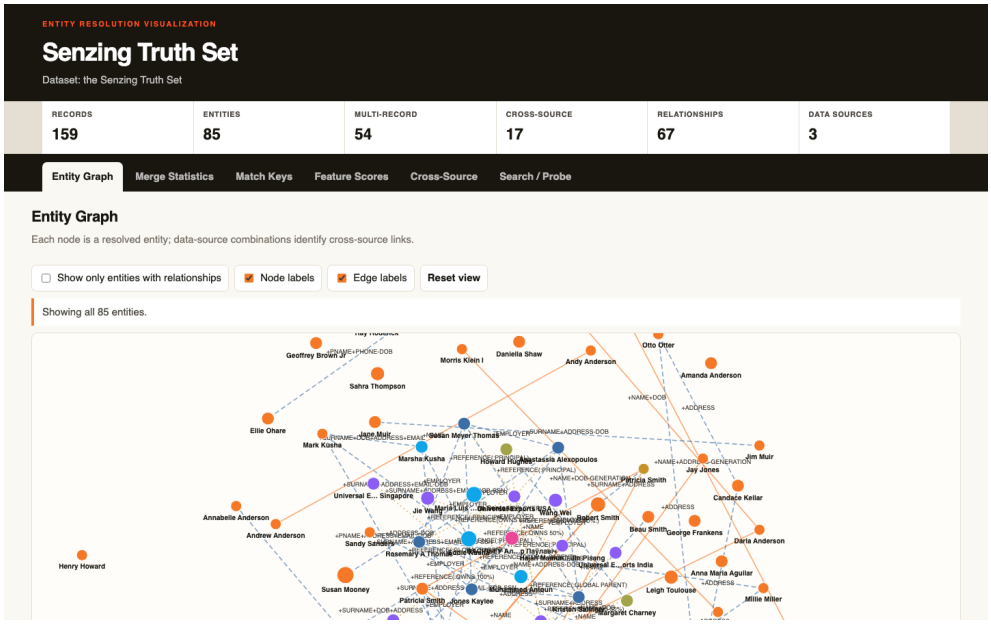
- The Senzing Truth Set is synthetic demo data, separate from the insurance-claims-fraud case data.
- The live visualization provides entity, relationship, statistics, search, and drilldown views; the saved HTML snapshot preserves the tabs backed by embedded data.
- The current visualization model reports 85 entities from 159 records, while the phase reference text states 84. This discrepancy remains documented but unresolved.

Questions & Responses

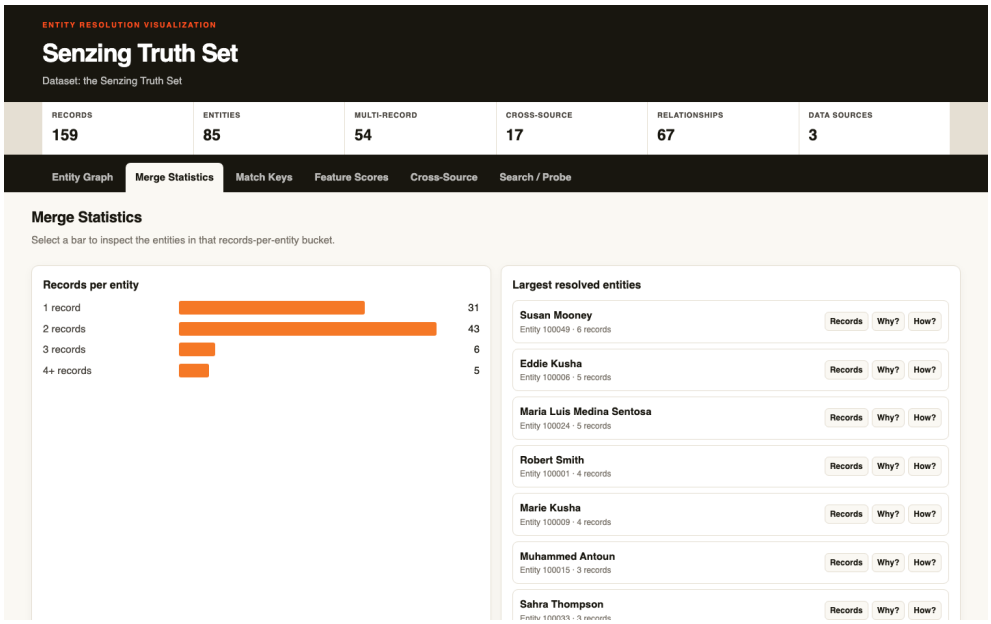
- **Q:** Would you like to switch to Opus 5.5 at high reasoning effort for this module?
  - **R:** No.
- **Q:** Are you ready to continue?
  - **R:** Yes; the bootcamper finished exploring.
- **Q:** Ready for me to stop the visualization server and clean up the Truth Set data?
  - **R:** Yes; stop the server and purge the Truth Set.

Actions Taken

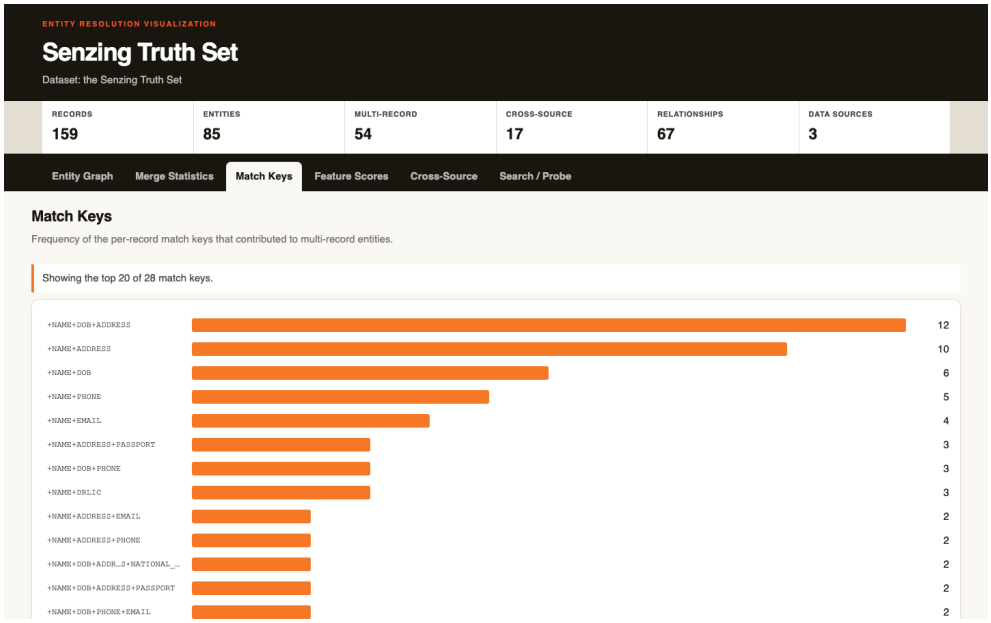
- Acquired, registered, and loaded 159 synthetic Truth Set records from CUSTOMERS, REFERENCE, and WATCHLIST; kept the insurance-case data unloaded.
- Built the Java visualization server and standalone snapshot at docs/visualizations/truthset_verification.html.
- Verified the live page, health check, and all 10 required API endpoints; each returned HTTP 200. The sample search returned 3 results.
- Observed 159 records, 85 entities, 54 multi-record entities, 17 cross-source entities, 67 relationships, and 3 data sources. The phase reference text states 84 entities; the discrepancy is recorded without guessing at its cause.
- Verified that all 4 source-combination legend rows match the 4 encoding keys; actual and synthetic collision counts were zero.
- Captured all 6 live tabs. The screenshot manifest flags the Entity Graph capture as unsettled; the other five captures completed without failures.
- Stopped server PID 935 and confirmed port 8080 was free. Deleted the 159 Truth Set records through the Java SDK and verified none remained.



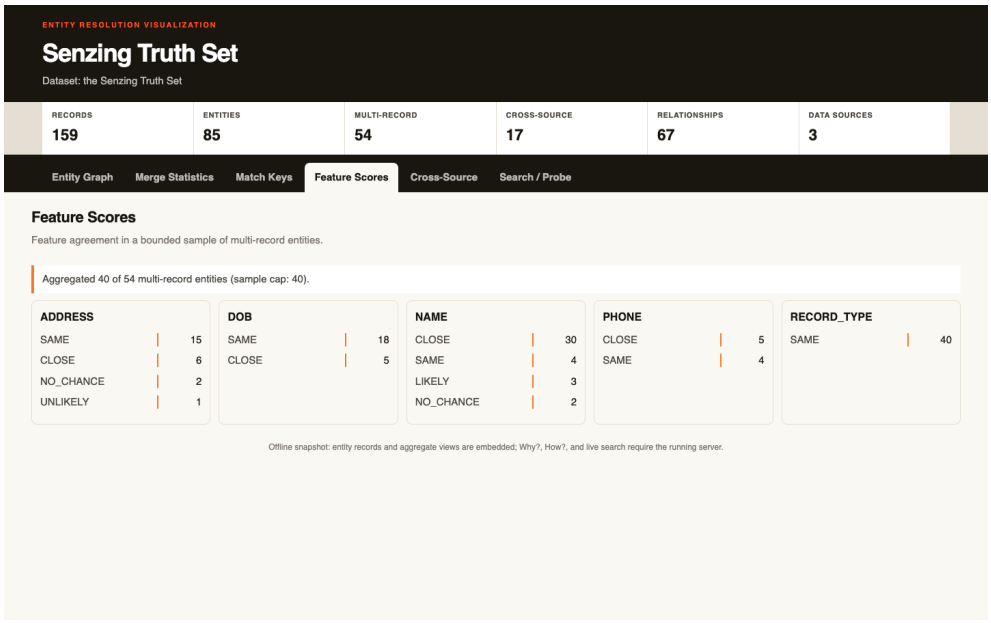
Entity Graph - 85 entities shown; headless layout unsettled



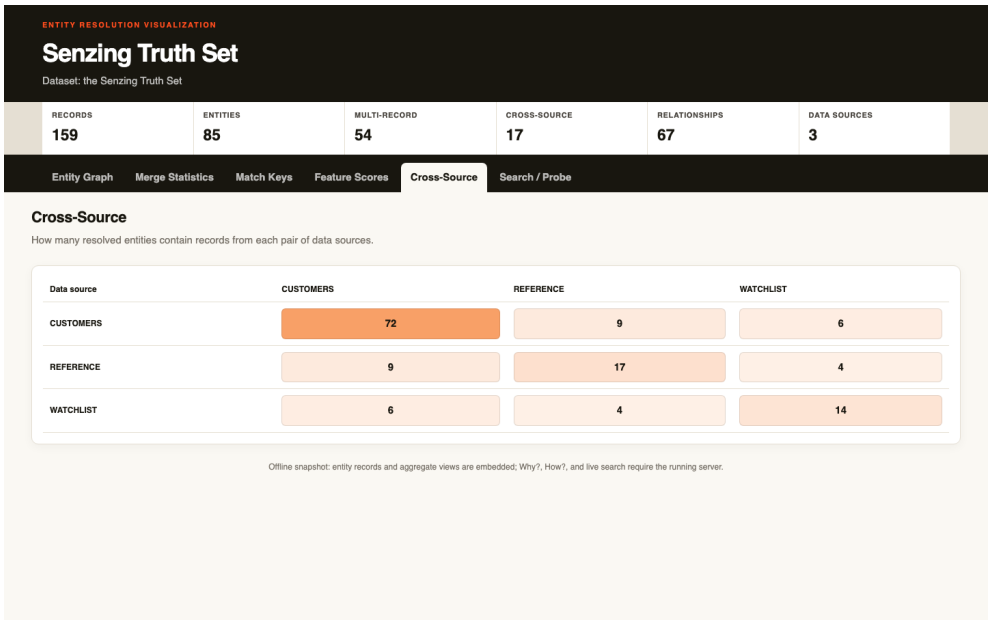
Merge Statistics - records-per-entity histogram and largest resolved entities



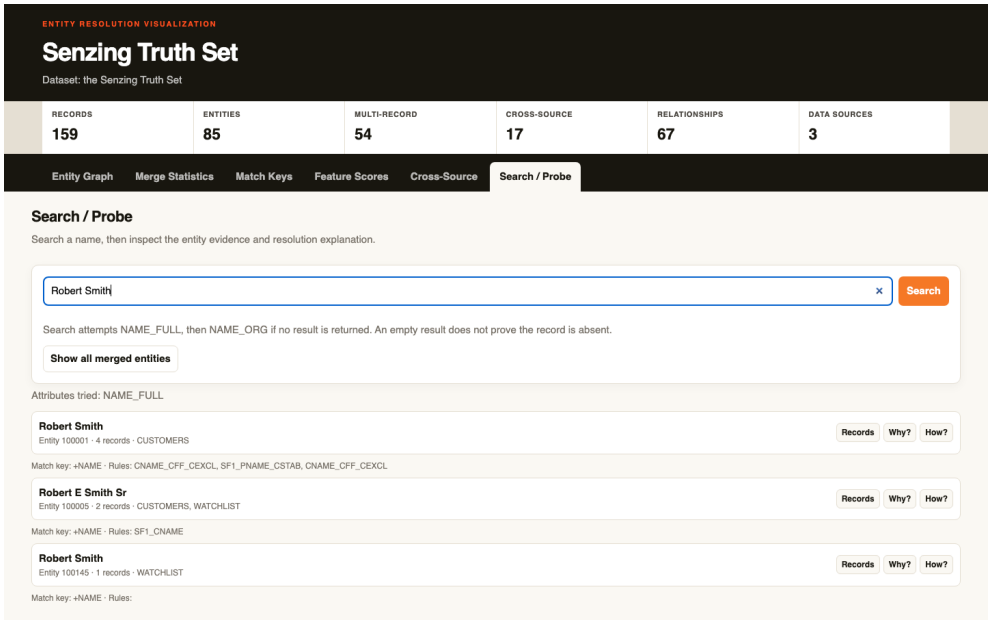
Match Keys - top 20 of 28 match keys by frequency



Feature Scores - sample of 40 multi-record entities across five features



Cross-Source - overlap counts across three data sources



Search / Probe - three entities returned by the sample query

End-of-Module Summary

What you accomplished:

- Loaded and explored the synthetic Truth Set in a live Java visualization, then stopped the server and removed all Truth Set records.
- Produced a standalone HTML snapshot and six screenshots covering the visualization tabs.
- Documented the unresolved 85-versus-84 entity-count difference and the unsettled Entity Graph screenshot.

Files produced:

- pom.xml - Java and JSON-P build configuration.
- src/server/TruthSetModel.java - entity model and visualization API payloads.
- src/server/SourceColorEncoder.java - source-set colors and encoding checks.
- src/server/TruthSetVizServer.java - loopback visualization server and API routes.
- src/server/TruthSetPage.java - generated six-tab visualization UI.
- src/server/d3.v7.min.js - vendored D3 asset for offline rendering.
- src/scripts/run-java.sh - Java launcher with Senzing runtime setup.
- src/system_verification/register_truthset_sources.java - Truth Set source registration.
- src/system_verification/load_truthset.java - Truth Set record loader.
- src/system_verification/purge_truthset_data.java - scoped SDK cleanup and verification.
- docs/visualizations/truthset_verification.html - standalone visualization snapshot.
- docs/visualizations/truthset_verification-tabs.json - screenshot coverage manifest.
- docs/visualizations/truthset_verification-entity-graph.png - Entity Graph capture.
- docs/visualizations/truthset_verification-merge-statistics.png - Merge Statistics capture.

- docs/visualizations/truthset_verification-match-keys.png - Match Keys capture.
- docs/visualizations/truthset_verification-feature-scores.png - Feature Scores capture.
- docs/visualizations/truthset_verification-cross-source.png - Cross-Source capture.
- docs/visualizations/truthset_verification-search-probe.png - Search / Probe capture.

Why it matters:

The visualization demonstrates entity resolution using synthetic data without loading the insurance case. The saved snapshot remains available offline, while the database is clean for the next module.

Data collection

Completed November 11, 2026

Information Shared

- The selected Senzing Demo Truth Set contains synthetic data and includes CUSTOMERS, REFERENCE, and WATCHLIST sources.
- Keep the collected files local and out of version control; reviewer approval remains pending before loading.

Questions & Responses

- **Q:** Would you like to switch to Sonnet 5.5 at medium reasoning effort for this module?
  - **R:** No.
- **Q:** How would you like to provide the data for this source? Reply with a number:
  - **R:** Use the Truth Set at https://github.com/Senzing/truth-sets/tree/main/truthsets/demo and asked whether the data could be found.
- **Q:** Should I use REFERENCE for VENDOR_REGISTRY and WATCHLIST for SIU_WATCHLIST?
  - **R:** Yes, use that mapping.

Actions Taken

- Downloaded and validated 120 CUSTOMERS records as POLICYHOLDERS, 22 REFERENCE records as VENDOR_REGISTRY, and 17 WATCHLIST records as SIU_WATCHLIST.
- Recorded each measured and expected count, file size, provenance, and validation result in config/data_sources.yaml.
- Confirmed the stored 500-record license limit covers the 159 collected records; no sample or license prompt was needed.
- Kept the data unloaded pending reviewer approval.

End-of-Module Summary

What you accomplished:

- Collected and validated 159 synthetic Truth Set records across the three approved business-source assignments.
- Documented source locations, validation status, collection status, and privacy handling; retained all records without loading them.

Files produced:

- data/raw/POLICYHOLDERS.jsonl - 120 validated CUSTOMERS records.
- data/raw/VENDOR_REGISTRY.jsonl - 22 validated REFERENCE records.
- data/raw/SIU_WATCHLIST.jsonl - 17 validated WATCHLIST records.
- config/data_sources.yaml - source registry with measured counts and validation checks.
- docs/data_source_locations.md - original source URLs and local file locations.

- docs/data_collection_checklist.md - inventory and validation checklist.
- docs/security_compliance.md - data-handling notes and reviewer-approval status.
- .gitignore - excludes raw source files from version control.

Why it matters:

The source files are documented and ready for quality evaluation, while remaining unloaded until reviewer approval.

Data Quality, Mapping, and Transformation

Completed November 11, 2026

Information Shared

- The quality assessment found all three sources below 70%, chiefly because of sparse field completeness. Missing values were not filled by inference.
- Improved inputs preserve the original collected files. Only documented, source-supported formatting changes were applied; ambiguous dates and meaningful identifier/contact differences were preserved.
- The user chose to use SIU_WATCHLIST as-is at 56.5%, with its quality limitation and one record without a name documented.

Questions & Responses

- **Q:** Would you like to switch to the recommended model and effort?
  - **R:** No; the current setting was retained.
- **Q:** Which source should be mapped first, and in what level of detail?
  - **R:** POLICYHOLDERS first, using concise mapping; then continued through VENDOR_REGISTRY and SIU_WATCHLIST.
- **Q:** How would you like to handle the below-threshold quality scores?
  - **R:** Initially chose to iterate on the data; after no further safe SIU_WATCHLIST corrections remained, chose to proceed with that source as-is.

Actions Taken

- Assessed 120 POLICYHOLDERS, 22 VENDOR_REGISTRY, and 17 SIU_WATCHLIST records; documented quality scores, field coverage, and distinct-value overlaps.
- Created improved source copies with safe formatting changes while preserving original files and record keys.
- Mapped all three sources, wrote Java transformers and per-source mapping specifications, and created load-ready outputs. The VENDOR_REGISTRY output also includes three deduplicated embedded employer records.
- Updated the source registry, evaluation report, progress state, mapping documentation, and transformation lineage. The optional sandbox validation was skipped; no records were loaded.

INSURANCE CLAIMS FRAUD · MODULE 5

Data quality assessment

Source coverage and observed format consistency before mapping. Quality scores help assess readiness to map; they are not measures of entity-match accuracy.

3	0 / 3	0
sources assessed	sources at or above 70%	duplicate source-key pairs

120 RECORDS · ORGANIZATION 6 · PERSON 114

POLICYHOLDERS

DATA_SOURCE in file: `CUSTOMERS`

51.2% quality score

Field completeness	33.2%
Format consistency	91.9%
Duplicate-free score (0.0% duplicate keys)	100.0%

Needs attention before mapping
The score is below 70%. This measures source completeness, observed formats, and duplicate keys—not match accuracy.

Per-field completeness

ADDR_CITY 38/120	31.7%
ADDR_COUNTRY 6/120	5.0%
ADDR_FULL 4/120	3.3%
ADDR_LINE1 69/120	57.5%
ADDR_POSTAL_CODE 37/120	30.8%
ADDR_STATE 34/120	28.3%
ADDR_TYPE 56/120	46.7%
DATE_OF_BIRTH 64/114	56.1%
DRIVERS_LICENSE_NUMBER 11/120	9.2%
DRIVERS_LICENSE_STATE 11/120	9.2%
EMAIL_ADDRESS 45/120	37.5%
GENDER 20/114	17.5%
NAME_FIRST 106/114	93.0%
NAME_FULL 6/120	5.0%
NAME_LAST 107/114	93.9%
NAME_MIDDLE 2/114	1.8%
NAME_ORG 6/6	100.0%
NAME_TYPE 1/120	0.8%
NATIONAL_ID_COUNTRY 3/120	2.5%
NATIONAL_ID_NUMBER 4/120	3.3%
PASSPORT_COUNTRY 15/120	12.5%
PASSPORT_NUMBER 15/120	12.5%
PHONE_NUMBER 21/120	17.5%
PHONE_TYPE 19/120	15.8%
RECORD_TYPE 120/120	100.0%
SSN_NUMBER 8/120	6.7%

Counts use person-only or organization-only applicability where appropriate; remaining fields are counted across all records. Structural keys are present in every record; unrecognized root fields are not included until mapping dispositions are decided.

Root fields for mapping: `AMOUNT` , `CATEGORY` , `DATE` , `STATUS`

22 RECORDS · ORGANIZATION 6 · PERSON 16

VENDOR_REGISTRY

DATA_SOURCE in file: `REFERENCE`

56.1% quality score

Field completeness	38.5%
Format consistency	96.6%
Duplicate-free score (0.0% duplicate keys)	100.0%

Needs attention before mapping
The score is below 70%. This measures source completeness, observed formats, and duplicate keys—not match accuracy.

Per-field completeness

ADDR_CITY 8/22	36.4%
ADDR_COUNTRY 6/22	27.3%
ADDR_FULL 7/22	31.8%
ADDR_LINE1 9/22	40.9%
ADDR_POSTAL_CODE 7/22	31.8%
ADDR_STATE 5/22	22.7%
ADDR_TYPE 15/22	68.2%
DATE_OF_BIRTH 2/16	12.5%
EMAIL_ADDRESS 4/22	18.2%
EMPLOYER 9/16	56.2%
LEI_NUMBER 2/6	33.3%
NAME_FIRST 2/16	12.5%
NAME_FULL 14/22	63.6%
NAME_LAST 2/16	12.5%
NAME_ORG 6/6	100.0%
NAME_TYPE 5/22	22.7%
NATIONAL_ID_COUNTRY 2/22	9.1%
NATIONAL_ID_NUMBER 2/22	9.1%
NATIONAL_ID_TYPE 2/22	9.1%
PHONE_NUMBER 8/22	36.4%
PHONE_TYPE 3/22	13.6%
RECORD_TYPE 22/22	100.0%
REL_ANCHOR_DOMAIN 4/22	18.2%
REL_ANCHOR_KEY 4/22	18.2%
REL_POINTER_DOMAIN 14/22	63.6%
REL_POINTER_KEY 14/22	63.6%
REL_POINTER_ROLE 14/22	63.6%
TAX_ID_COUNTRY 4/22	18.2%
TAX_ID_NUMBER 4/22	18.2%
TAX_ID_TYPE 4/22	18.2%

Counts use person-only or organization-only applicability where appropriate; remaining fields are counted across all records. Structural keys are present in every record; unrecognized root fields are not included until mapping dispositions are decided.

Root fields for mapping: `CATEGORY` , `DATE` , `STATUS`

17 RECORDS · PERSON 17

SIU_WATCHLIST

DATA_SOURCE in file: `WATCHLIST`

56.2% quality score

End-of-Module Summary

What you accomplished:

- Assessed and mapped all three insurance-fraud data sources, retaining documented quality limitations.
- Prepared validated JSONL outputs for later processing without loading records in this module.

Files produced:

- data/senzing-ready/POLICYHOLDERS.jsonl - mapped POLICYHOLDERS output.
- data/senzing-ready/VENDOR_REGISTRY.jsonl - mapped vendors and embedded employers.
- data/senzing-ready/SIU_WATCHLIST.jsonl - mapped SIU_WATCHLIST output, used as-is.
- docs/data_source_evaluation.md - source quality scores, coverage, overlaps, and limitations.
- docs/mapping/mapping_POLICYHOLDERS.md, docs/mapping/mapping_VENDOR_REGISTRY.md, and docs/mapping/mapping_SIU_WATCHLIST.md - source transformation guides.
- docs/mapping/POLICYHOLDERS_mapper.md, docs/mapping/VENDOR_REGISTRY_mapper.md, and docs/mapping/SIU_WATCHLIST_mapper.md - per-source mapping specifications.
- src/transform/transform_POLICYHOLDERS.java, src/transform/transform_VENDOR_REGISTRY.java, and src/transform/transform_SIU_WATCHLIST.java - Java transformers.
- src/transform/JsonSupport.java - shared JSON support.
- config/data_sources.yaml and config/bootcamp_progress.json - source registry and bootcamp progress.
- docs/visualizations/data_quality_assessment.html - offline quality visualization.
- docs/visualizations/data_quality_assessment.png - quality visualization screenshot.

Why it matters:

The mapped outputs are documented and ready for the next processing stage, while the low-completeness scores and missing-name gap remain visible and no source records have been loaded here.

Data processing

Completed November 11, 2026

Information Shared

- All three sources loaded in full: POLICYHOLDERS 120, VENDOR_REGISTRY 25, and SIU_WATCHLIST 17; the mapped/load total is 162.
- The VENDOR_REGISTRY count differs from the 22 collected records because its mapping produces three deduplicated embedded employer organizations.
- SDK validation found 90 entities, including 15 cross-source entities; all 52 multi-record entities passed the how-state check.
- Sampled match keys and relationships include suppressors; relationship-level conflicts describe candidate links that were not resolved together, not a match-accuracy score.
- The fictional scenario has no real stakeholders. UAT was self-directed, and the sample findings are not ground-truth labels; precision, recall, F1, and absolute match accuracy are unavailable.

Questions & Responses

- **Q:** Are you ready to move on to Data processing?
  - **R:** Yes.
- **Q:** Would you like to switch to the recommended model and effort for this code-heavy module?
  - **R:** No; kept the current setting.
- **Q:** Which production-volume tier should guide the loading program?
  - **R:** Medium.
- **Q:** Is reviewer approval confirmed for the test and full loads?
  - **R:** Yes.
- **Q:** Which loading strategy should production use?
  - **R:** Incremental.
- **Q:** Is the inventory of data sources complete?
  - **R:** Yes; POLICYHOLDERS, VENDOR_REGISTRY, and SIU_WATCHLIST.
- **Q:** Are there dependencies that require a particular source load order?
  - **R:** No.
- **Q:** Which orchestration approach should be used for the remaining sources?
  - **R:** Sequential.

Actions Taken

- Loaded 25 VENDOR_REGISTRY records; the 22-to-25 difference is reconciled to three embedded employer organizations. Processed 54 redo records and emptied the queue.
- Sequentially loaded 17 SIU_WATCHLIST and 120 POLICYHOLDERS records, with no failures or retryable records; final SDK record count was 162. Processed 5 additional redo records and emptied the queue.
- Created and ran src/load/ValidateEntityResults.java; the SDK export and record-ID queries reconciled to

162 records and 90 entities.

- Reviewed all 15 cross-source entities and 8 single-source multi-record samples. Eight cross-source samples had corroborating evidence; seven require human follow-up. These are provisional assessments, not truth labels.
- Audited 47 single-source and 24 cross-source per-record match keys, 70 deduplicated relationships, and all 52 multi-record entity how-states. Cross-source per-record suppressors: 0; unsettled how-states: 0.
- Recorded load statistics, proxy UAT findings, the absence of real stakeholders, and the lack of ground-truth accuracy metrics.

End-of-Module Summary

What you accomplished:

- Loaded and reconciled all three sources, then drained redo processing.
- Reviewed cross-source matches and single-source examples, documenting provisional follow-up findings without claiming absolute accuracy.

Files produced:

- src/load/ValidateEntityResults.java - SDK-backed entity, relationship, match-key, and how-state validation.
- logs/entity_validation.json - validation metrics and sampled entity evidence.
- docs/uat_results.md - self-directed UAT results and review candidates.
- docs/results_validation.md - reconciled load, entity, relationship, and audit results.
- docs/stakeholder_summary_module6.md - executive summary of outcomes, risks, and recommendations.
- docs/loading_strategy.md - final load reconciliation and validation findings.

Why it matters:

The project now has a reconciled processing baseline and specific cases for review. The synthetic watchlist links remain leads for investigation, not fraud conclusions, and measured accuracy requires adjudicated labels.

Query, Visualize and Discover

Completed November 12, 2026

Information Shared

- Derived four query requirements from the refined insurance-claims-fraud criteria: policyholder deduplication, evidence-supported vendor connections, watchlist review leads, and a master list/API/report output.
- Explained that 162 loaded records resolved into 90 entities, and that entity counts and possible-match indicators are review signals rather than truth labels. Absolute accuracy remains unmeasured without adjudicated labels.
- Presented the Why/How evidence model, entity networks, and a two-hop path through disclosed source relationships.
- Reviewed source overlap measurements and the distinction between per-record and relationship suppressors.

Questions & Responses

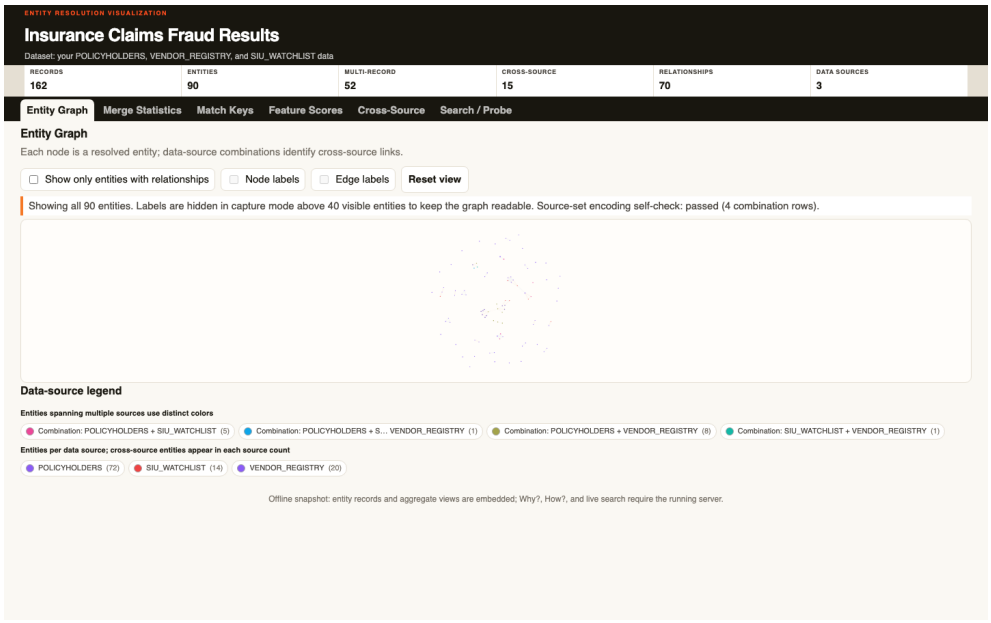
- **Q:** Would you like to switch to Claude Opus 5.5 for this module?
  - **R:** No.
- **Q:** Is there anything you'd like to adjust?
  - **R:** No; the derived requirements are accurate.
- **Q:** Would you like an interactive visualization of your resolved data - entity graph with its relationship view, merge statistics, cross-source overlap, and match/feature analysis, all in one app?
  - **R:** Yes.
- **Q:** Would you like to explore Senzing's advanced discover capabilities using examples from your own data?
  - **R:** Yes.
- **Q:** What would you like to do next? Reply with a number:
  - **R:** 1 - continue to How Analysis.
- **Q:** What would you like to do next? Reply with a number:
  - **R:** 1 - continue to Relationship Networks.
- **Q:** Would you like to continue to module completion now?
  - **R:** Yes.
- **Q:** Ready for me to stop the visualization server?
  - **R:** Yes.

Actions Taken

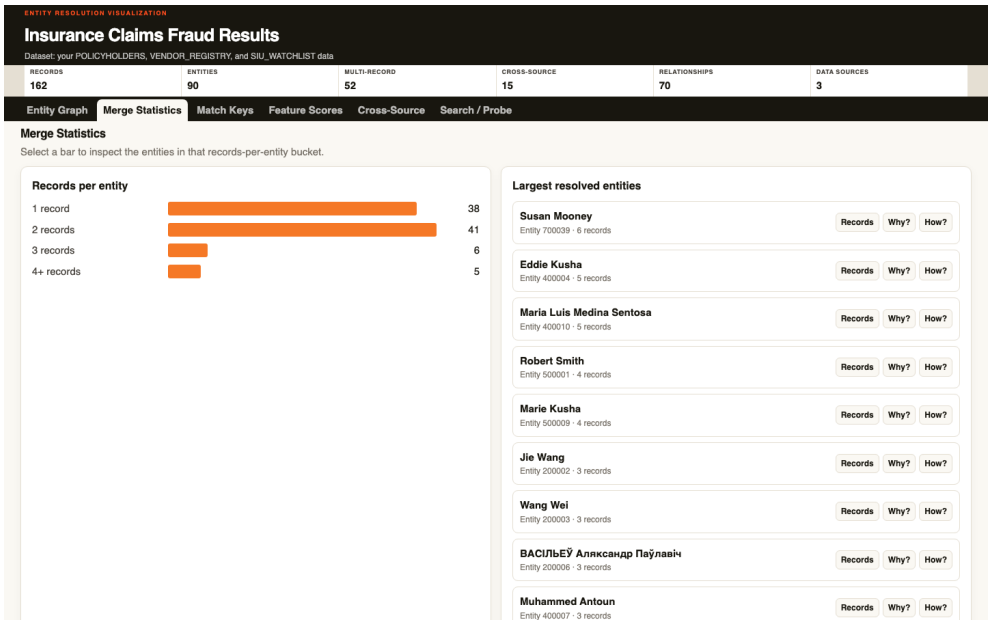
- Created and ran Java query programs for the master list, source-specific searches, quality samples, cross-source review queue, merge audit, and relationship networks. Documented the programs in docs/query_programs.md.
- Built and verified the six-tab visualization. Its offline snapshot retains the aggregate views; live Why, How,

and Search require the server.

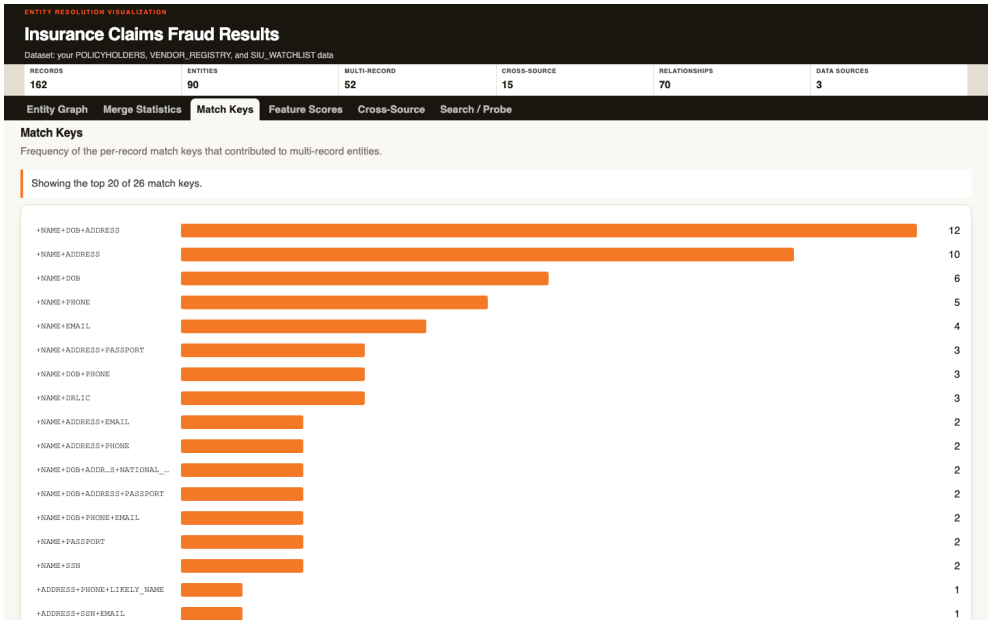
- Completed Discover steps 4a-4d, including Why and How analysis for Entity 400004 and a two-hop relationship path through Entity 200007.
- Produced the data-discoveries Markdown and PDF. Extracted PDF text was verified to contain the findings.
- Stopped visualization server PID 27637 with approval and confirmed port 49377 no longer served requests.



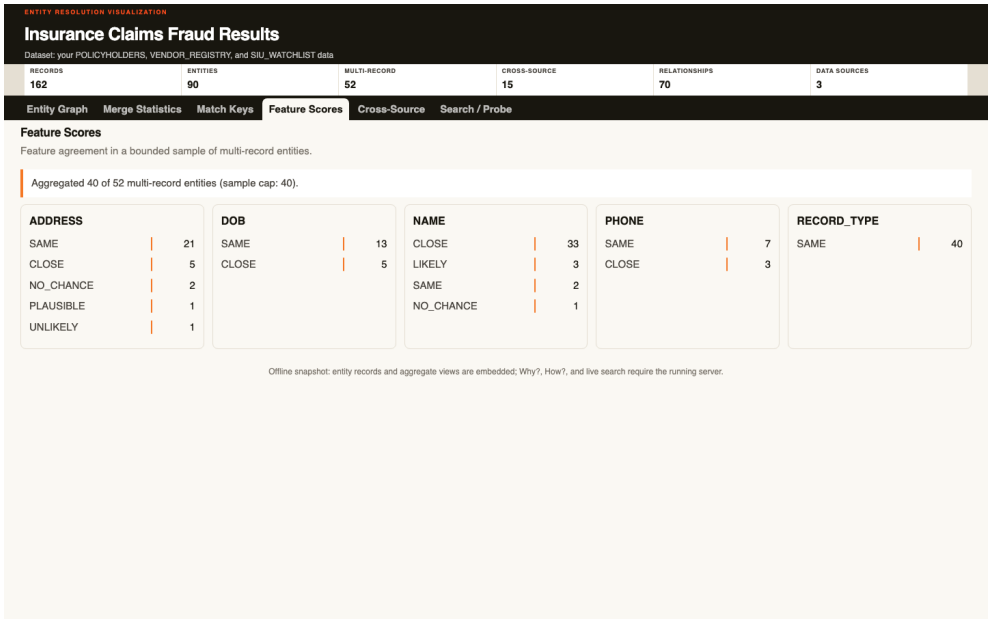
Entity Graph showing 90 entities and the source-combination legend



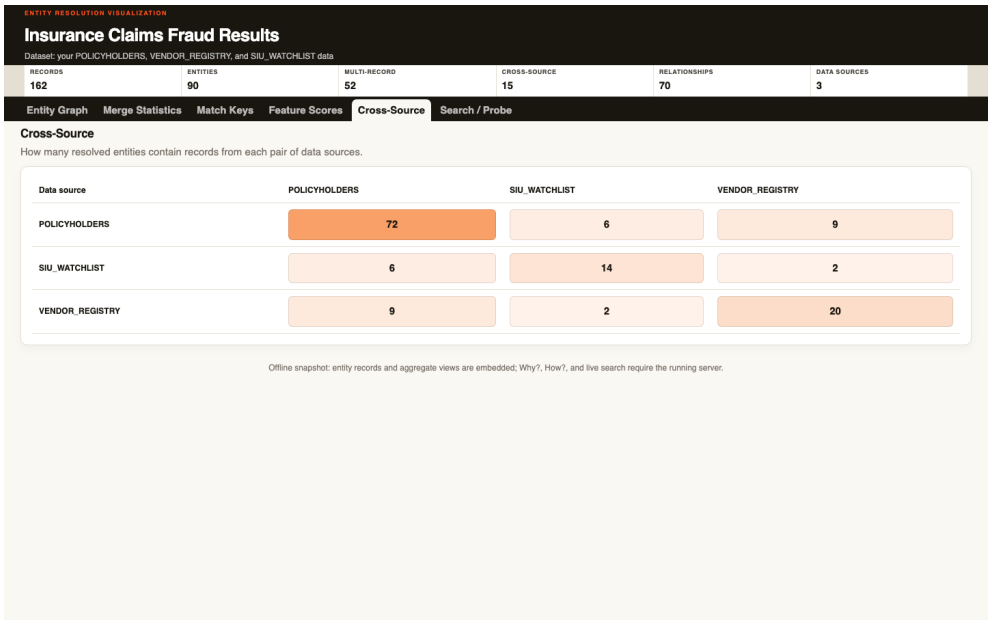
Merge Statistics showing records-per-entity buckets and largest entities



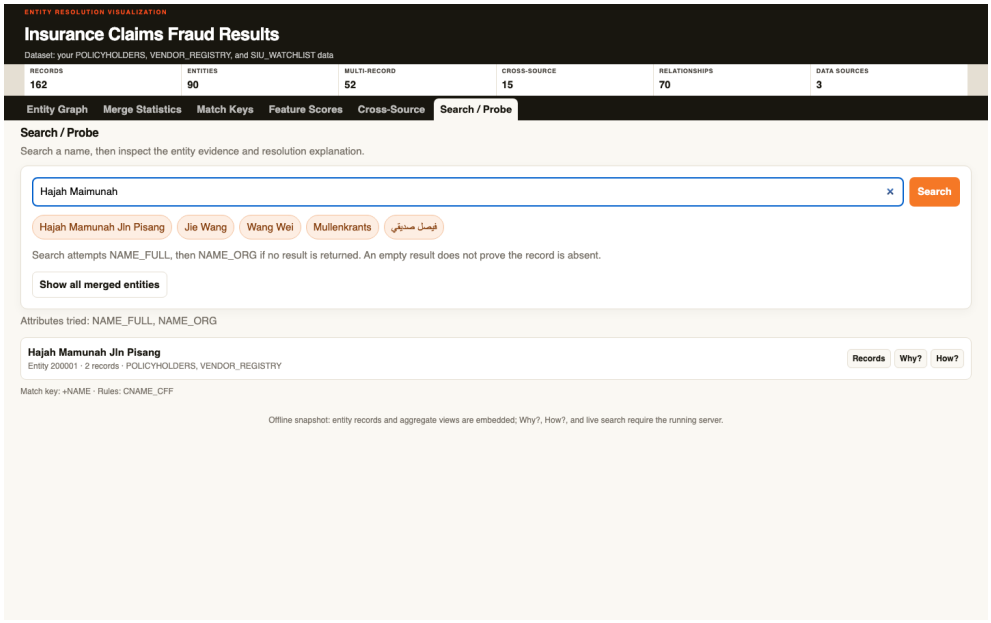
Match Keys showing the top 20 of 26 per-record keys



Feature Scores aggregated across a 40-entity sample



Cross-Source matrix of entity counts by source pair



Search / Probe for Hajah Maimunah with a result card

End-of-Module Summary

What you accomplished:

- Created and tested Java query programs, then explored resolution evidence, entity patterns, and relationships.
- Reviewed 14 cross-source POSSIBLY_SAME pairs and documented all 72 How resolution steps across 52 multi-record entities.
- Built the interactive results visualization and completed the Discover phase.

Files produced:

- docs/bootcamp_data_discoveries.md - evidence-backed findings, merge audit, review queue, and overlap measurements.
- docs/bootcamp_data_discoveries.pdf - verified PDF version of the discoveries report.
- src/query/BuildCrossSourceReviewQueue.java - SDK-backed cross-source review queue builder.
- src/query/BuildHowMergeAudit.java - SDK-backed How resolution-step audit.
- docs/query_programs.md - query program usage and output documentation.
- reports/query/cross_source_review_queue.json - 14 cross-source review candidates.
- reports/query/how_merge_audit.json - 72 resolution steps with their match keys.
- reports/query/relationship_networks.json - network and two-hop path evidence.
- docs/visualizations/results_visualization.html - offline results-visualization snapshot.
- docs/visualizations/results_visualization-entity-graph.png - captured Entity Graph tab.
- docs/visualizations/results_visualization-merge-statistics.png - captured Merge Statistics tab.
- docs/visualizations/results_visualization-match-keys.png - captured Match Keys tab.
- docs/visualizations/results_visualization-feature-scores.png - captured Feature Scores tab.
- docs/visualizations/results_visualization-cross-source.png - captured Cross-Source tab.

- docs/visualizations/results_visualization-search-probe.png - captured Search / Probe tab.

Why it matters:

The query evidence and review queue support earlier investigation of potential claims-fraud connections without treating a match as a fraud verdict. The source-overlap findings contextualize the results, while the lack of adjudicated labels keeps absolute accuracy unmeasured.

Senzing

SENZING BOOTCAMP

Certificate of Completion

Entity Resolution, from first principles to a production pipeline

THIS CERTIFICATE IS PROUDLY PRESENTED TO

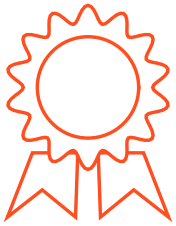
Senzing Bootcamper

for successfully completing 9 modules of the Senzing Bootcamp - Senzing's guided path from defining the business problem to a working entity-resolution pipeline:

Entity Resolution Concepts · Discover the Business Problem · SDK setup · System verification · Truth Set visualization · Data collection · Data Quality, Mapping, and Transformation · Data processing · Query, Visualize and Discover

November 12, 2026

DATE COMPLETED



Senzing Bootcamp

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